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Evaluating the potential of profiling users from visual behavior data in pedestrian navigation

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ABSTRACT

Recognizing user attributes during wayfinding is essential for developing adaptive and personalized navigation aids. Machine learning models trained on gaze data offer a promising approach, yet most prior studies have been conducted in controlled laboratory settings, leaving their applicability and interpretability in dynamic real-world contexts unclear. This study investigates the potential of eye movement data for user profiling in both immersive virtual reality and real-world environments. Using machine learning techniques, we analyzed gaze behavior to recognize five user attributes: gender, expertise, spatial ability, environmental familiarity, and navigation task. A comprehensive leave-one-X-out validation (X = task, person, environment) was adopted to evaluate model generalizability across tasks, users, and environments. The Light Gradient-Boosting Machine (LightGBM) achieved the best overall performance, with task recognition consistently outperforming other attributes. However, generalizability declined for unseen users and environments. Statistical analyses further showed that basic statistical features, pupil dilation measures, and fixation semantic features were particularly informative for user profiling. These findings highlight both the potential and current limitations of gaze-based user modeling for personalized navigation in real-world applications.

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KEYWORDS

User modeling; pedestrian navigation; machine learning; wayfinding; eye tracking

1. Introduction

Navigation has long been a fundamental human activity, but with increasing human mobility and urbanization, it has become an even more integral part of daily life for urban residents and visitors (Munio et al., 2019; Weisberg et al., 2014). As a cognitively demanding task, it requires spatial self-awareness and sequential decision-making (Montello, 2005; Stern & Portugali, 1999). These demands have driven the widespread adoption of navigation aid systems and a shift from device-centered to human-centered design, emphasizing adaptability and personalization (Kang, 2025; Torkamaan et al., 2024). A key aspect of this shift is the automatic adaptation of navigation aids based on user attributes such as gender, spatial ability, task type, expertise, environmental familiarity, cognitive load and personality (Griffin et al., 2024; Huang et al., 2018; Keskin et al., 2020; Ooms et al., 2014; Raubal & Panov, 2009). As these attributes closely relate to users' preferences, motivations and information needs, inferring them for user profiling is essential to developing adaptive navigation systems.

User attributes can be inferred from various data sources, including geographic big data (Liu et al., 2025; Wu et al., 2019), social media data (Bi et al., 2013; Kröger et al., 2022; Yu et al., 2019), electroencephalographic (EEG) signals (Honal & Schultz, 2005), and genome information (Excoffier et al., 2013). This study focuses specifically on visual behavior data, as empirical evidences shows that individuals with different attributes exhibit distinct eye movement patterns, which can be leveraged for attribute inference (Figure 1) (Harms et al., 2021; Liao & Dong, 2017; Siła-Nowicka et al., 2016). For instance, geography experts demonstrate higher fixation frequency than non-experts during map-based spatial orientation tasks (Dong et al., 2018; Ooms et al., 2014), while females tend to display more exploratory gaze behaviors (e.g. longer scan paths) when viewing natural scenes (Abdi Sargezeh et al., 2019). Eye movement data provide unique insights into users' cognitive processes and convey rich biometric information that reflects individual user attributes (Aziz et al., 2024).

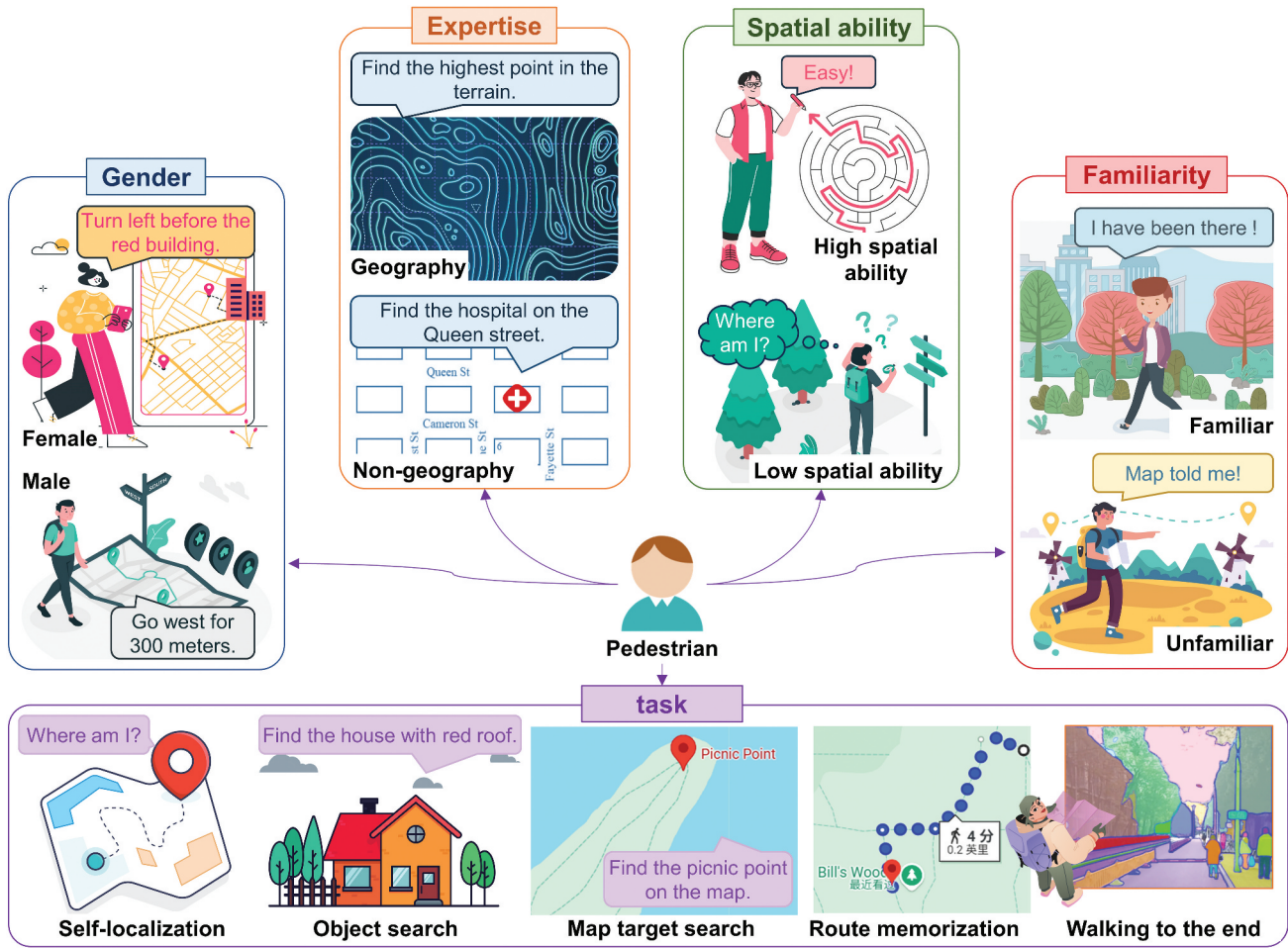


Figure 1. User attributes and navigation behavior.

User attribute recognition is fundamentally a classification problem (e.g. binary classification: male vs. female) and can be effectively addressed using machine learning algorithms. Studies across psychology, education and computer science have demonstrated the feasibility of inferring user attributes from eye movement features. A selection of these studies is summarized in Appendix Table A1. While existing research has shown promising results in recognizing individual attributes, substantial gaps remain in applying these methods to pedestrian navigation contexts, which are elaborated below.

- (1) **Ecological validity.** Compared to image viewing and text reading tasks in controlled laboratory settings (e.g. Al Zaidawi et al., 2020; Barral et al., 2020; Henderson et al., 2013; Steil et al., 2019; Vallabh Varsha Haria et al., 2022), profiling users during navigation in geographic environments presents additional challenges. These stem from the spatial

complexity of such environments and the dynamic, context-dependent nature of human navigation behavior (Yang et al., 2025). While some studies have explored wayfinding behavior and attribute recognition in (immersive) virtual reality (VR) environments (Gao et al., 2023; Gil-López et al., 2023; Orlosky et al., 2019; Snopková et al., 2019, 2022; Stachoň et al., 2025) and real-world settings (Liao, Dong, et al., 2019), extending the findings from fixed and simplified laboratory experiments to large-scale and complex geographic environments remains underexplored.

- (2) **Generalizability.** An adaptive navigation aid system should function effectively across new contexts, including new users, tasks, and environments, meaning that the user attribute recognition model maintains high or acceptable performance when applied to data distributions not encountered during training. However, existing model evaluation methods,

such as randomly splitting data into training and test sets (e.g. K-fold cross-validation), may not adequately assess model generalizability (Abdi Sargezeh et al., 2019; Barral et al., 2020; Li et al., 2020). To address this, we employ a comprehensive Leave-One-X-Out (LOXO) cross-validation, where X denotes person, task, or environment (e.g. roads). This approach offers a more rigorous evaluation of model performance on unseen data but remains underutilized in existing research (Celiktutan & Demiris, 2019; Liao, Dong, et al., 2019).

- (3) **Explainability.** The explainability of machine learning models in user attribute recognition is crucial for enhancing transparency and interpretability (Gilpin et al., 2018). While previous studies have demonstrated the feasibility of using eye movement features for user attribute classification, limited research has explored which specific features contribute most and why they are cognitively relevant. A crucial step toward explainability involves quantitatively assessing feature importance in classification models (Gao et al., 2023; Kattenbeck et al., 2025; Liao et al., 2022). Beyond this, statistical hypothesis testing can validate whether specific eye movement features significantly differ between user groups, thus providing empirical evidence for cognitive-behavioral connections and distinctions, but has been addressed by only a few studies (Barral et al., 2020; Gil-López et al., 2023).

Motivated by these gaps, this study investigates user profiling based on eye movement data collected during pedestrians navigate in two different environments: an immersive virtual reality (VR) indoor environment (Study 1, Section 2) and a real-world (RE) outdoor environment (Study 2, Section 3). We trained and tested a LightGBM model (Ke et al., 2017) to recognize five user attributes: gender, expertise, spatial ability, environmental familiarity, and task. Specifically, we aim to address the following three research questions (RQs):

- (1) **RQ1 (Feasibility):** Can gaze behavior collected during pedestrian navigation in immersive VR and real-world settings reliably predict user attributes?
- (2) **RQ2 (Generalizability):** How generalizable are gaze-based user recognition models across different users, tasks, and environments?

- (3) **RQ3 (Interpretability):** Which eye movement features contribute most to user attribute recognition, and how do they explain cognitive differences among users?

This study contributes by (1) extending gaze-based user profiling to immersive VR and real-world navigation contexts (RQ1), (2) systematically evaluating model generalizability through K-Fold (baseline) and LOXO cross-validations (RQ2), and (3) improving model explainability through feature importance analysis and statistical testing (RQ3).

2. Study 1: indoor navigation in virtual reality

2.1. Overview

Study 1 investigates user attribute recognition using basic eye movement features. The data originated from an indoor navigation study conducted by Dong et al. (2022) in an immersive VR environment, modeled after a university building. Forty participants ($N = 40$) navigated the environment using a joystick while wearing an HTC VIVE head-mounted display (HMD) equipped with a 90 Hz eye tracker. The Santa Barbara Sense of Direction Scale (SBSODS) (Hegarty et al., 2002) was used to measure participants' spatial abilities, which is particularly relevant given the direction-sensitive nature of the navigation tasks. Each participant completed four tasks: one *free-viewing* task and three *wayfinding* tasks, resulting in a total of 160 task trials ($40 \text{ participants} \times 4 \text{ tasks}$). They granted written consent and were compensated for participation. The experiment was reviewed and approved by the local Institutional Review Board. For more experimental details, refer to Dong et al. (2022).

This study considered the following five user attributes:

- (1) **Gender.** The experiment contained 30 females and 10 males.
- (2) **Expertise.** Participants came from a range of disciplines, including geography, psychology, computer science, arts, and engineering. Previous research indicates that geographical education influences students' spatial abilities and visual behaviors (Dong et al., 2018, 2023). Based on this, we categorized participants into two groups: Geography (Geo) ($N = 30$) and Non-geography (Non-geo) ($N = 10$).
- (3) **Familiarity.** Environmental familiarity was determined by simply asking participants whether they had visited the building before.

Nine participants have never visited the building and thus were labeled as “Unfamiliar” ($N=9$). The remaining 31 participants had visited the building and were labeled as “Familiar” ($N=31$).

- (4) **Spatial ability.** The mean SBSODS score of the participants was 61.95 ($SD=20.21$). We binarized spatial ability using the median value (60.5), meaning that a participant was assigned either a “high” (≥ 60.5 , $N=20$) or “low” (< 60.5 , $N=20$) level of spatial ability (Sun et al., 2020).
- (5) **Task.** We labeled each task with either “free-viewing” ($N=10$) or “wayfinding” ($N=30$) for binary classification.

2.2. Feature extraction

We extracted eye movement features for each participant and task trial. For fixation duration, saccade duration, saccade amplitude, and saccade average velocity, we computed 7 statistics: mean, minimum (min), maximum (max), standard deviation (std), 1/4 quartile ($q0.25$), median, and 3/4 quartile ($q0.75$), resulting in 28 features. Additionally, we computed the following 6 features: fixation frequency (per second), saccade frequency (per second), total fixation duration of a trial (FixTime), the time interval between the first and last fixation of a trial (FixTestTime), total saccade duration of a trial (SacTime), and the time interval between the first and last saccade (SacTestTime). Therefore, a total of 34 features were extracted. In Study 1, we focused on these basic and scene-independent gaze features to test the predictive range of common gaze dynamics in VR settings.

2.3. Sample balancing

As outlined in Section 2.1, the sample sizes are imbalanced, except for spatial ability. This imbalance may

lead to a bias toward the majority class in certain probability-based models, such as decision trees (Hanskunatai, 2018; Tyagi & Mittal, 2020). To address the issue, we chose the Synthetic Minority Oversampling Technique (SMOTE) (<https://imbalanced-learn.org/stable/>) (Lemaitre et al., 2017) to increase the number of samples in minority classes for gender, expertise, familiarity, and task. Oversampling was applied only to the training set to avoid bias in the testing set.

2.4. Classifier selection

We conducted a pilot study to evaluate 11 commonly used classifiers for identifying the five user attributes. All classifiers were tested using default hyperparameters and a 10-fold cross-validation method. Performance was measured using mean accuracy (ACC) and F1-score (F1), as defined in Equation (1) and (2).

$$accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

$$F1 = \frac{2TP}{2TP+FP+FN} \quad (2)$$

where TP, TN, FP, FN represent true positive, true negative, false positive and false negative, respectively.

The comparison results are presented in Table 1. Random Forest classifier achieved the highest accuracy and F1-score for recognizing gender ($ACC=0.819$, $F1=0.877$) and spatial ability ($ACC=0.725$, $F1=0.722$). Meanwhile, LightGBM outperformed other classifiers in recognizing expertise ($ACC=0.769$, $F1=0.844$), familiarity ($F1=0.844$), and task ($ACC=0.875$, $F1=0.907$). Given its superior performance, we selected LightGBM for subsequent analysis. LightGBM addresses the inefficiency and limited scalability of traditional Gradient Boosting Decision Tree (GBDT) models when handling large

Table 1. The performance of 11 classifiers for recognizing five user attributes in Study 1. Mean accuracy (ACC) and mean F1-score of 10 folds are presented. The bold values represent the best.

Classifier	Gender		Expertise		Spatial ability		Familiarity		Task	
	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
GBoost	0.781	0.851	0.750	0.833	0.725	0.691	0.731	0.824	0.856	0.896
LightGBM	0.706	0.797	0.769	0.844	0.681	0.670	0.750	0.844	0.875	0.907
MLP	0.806	0.859	0.700	0.792	0.713	0.678	0.713	0.810	0.769	0.837
AdaBoost	0.688	0.781	0.688	0.786	0.719	0.702	0.756	0.834	0.869	0.907
DecisionTree	0.750	0.826	0.681	0.783	0.625	0.608	0.713	0.791	0.788	0.854
LDA	0.700	0.769	0.563	0.666	0.631	0.611	0.550	0.677	0.781	0.840
KNN	0.650	0.714	0.625	0.702	0.713	0.704	0.575	0.640	0.719	0.788
LR	0.700	0.769	0.525	0.641	0.619	0.626	0.600	0.711	0.813	0.867
SVC	0.781	0.849	0.694	0.786	0.719	0.698	0.681	0.782	0.788	0.853
XGBoost	0.763	0.836	0.725	0.807	0.706	0.707	0.738	0.832	0.819	0.877
RF	0.819	0.877	0.738	0.827	0.725	0.722	0.744	0.836	0.838	0.882

Notes: MLP = multilayer perceptron; LDA = linear discriminant analysis; KNN = k-nearest neighbors; LR = logistic regression; SVC = support vector classifier; RF = random forest.

datasets or high-dimensional feature spaces by incorporating two key techniques: Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). GOSS prioritizes data instances with larger gradients when estimating information gain, enabling accurate approximation using fewer samples. EFB combines mutually exclusive features to reduce feature dimensionality without substantially compromising split accuracy. As a result, LightGBM achieves training speeds several times faster than conventional GBDT models while maintaining comparable accuracy (Ke et al., 2017). To further optimize the classifier's performance, we employed a grid search to identify the optimal hyperparameters for each user attribute.

2.5. Cross-validation methods

We employed the following three cross-validation methods to perform user attribute recognition.

- (1) **K-Fold (Baseline method).** K-fold cross-validation is a widely used technique in machine learning for model training and evaluation (Wong and Yeh, 2019). In this study, we implemented 10-fold cross-validation ($K = 10$), where the dataset was randomly partitioned into training and testing sets. Each attribute underwent 10-fold validation, resulting in a total of 50 runs ($5 \text{ attributes} \times 10 \text{ folds}$).
- (2) **Leave-One-Task-Out (LOTO).** In LOTO, data from one task were extracted as the testing set, while the remaining tasks formed the training set. This process was repeated until each task had been tested once. LOTO evaluated the model's ability to generalize user attribute recognition across new tasks. As task recognition was not applicable in this setting, LOTO was conducted only for the remaining four attributes, resulting in 16 runs ($4 \text{ attributes} \times 4 \text{ tasks}$).
- (3) **Leave-One-Person-Out (LOPO).** In LOPO, data from one participant were used as the testing set, while the data from the remaining participants formed the training set. This process continued until each participant had been tested once. LOPO was to assess the model's ability to generalize attribute recognition to new users. Given five attributes and 40 participants, this method resulted in 200 runs ($5 \text{ attributes} \times 40 \text{ participants}$).

2.6. Results

2.6.1. Performance across three cross-validation scenarios

Figure 2 presents the mean accuracy and F1-score of the LightGBM model across three cross-validation

scenarios. All accuracies exceed the chance level (0.5), demonstrating the model's effectiveness in attribute recognition in VR.

In K-fold cross-validation, task recognition achieved the highest performance ($ACC = 0.863$, $F1 = 0.903$), followed by gender ($ACC = 0.794$, $F1 = 0.860$), familiarity ($ACC = 0.731$, $F1 = 0.819$), and expertise ($ACC = 0.719$, $F1 = 0.813$). Spatial ability exhibited the lowest recognition accuracy ($ACC = 0.656$, $F1 = 0.659$).

In LOTO validation, the model demonstrated comparable performance to K-fold in recognizing gender ($ACC = 0.756$, $F1 = 0.837$), but outperformed K-fold in identifying expertise ($ACC = 0.775$, $F1 = 0.852$), spatial ability ($ACC = 0.681$, $F1 = 0.692$), and familiarity ($ACC = 0.775$, $F1 = 0.858$).

In contrast, LOPO validation yielded lower performance than K-fold for all attributes except task recognition, which surpassed K-fold with an accuracy of 0.881 ($F1 = 0.912$). The mean accuracies for the remaining attributes in LOPO were: gender ($ACC = 0.638$, $F1 = 0.625$), spatial ability ($ACC = 0.606$, $F1 = 0.366$), and expertise = familiarity ($ACC = 0.600$, $F1 = 0.633$).

2.6.2. Feature importance and statistical analysis

As a tree-based model, LightGBM tracks the number of times each feature is used in model splits, providing a measure of feature importance in classification (Qin et al., 2024). We recorded feature importance across all classification runs and ranked the top five most important features. Given the variation in top-ranked features across the three cross-validation methods and classification runs, averaging importance scores would be inappropriate. To ensure a rigorous assessment, we employed a comprehensive ranking method in which the five highest-ranked features from each cross-validation method were selected and combined into a feature set, resulting in a maximum of 15 dimensions. These selected features were then ranked based on their cumulative importance, with lower cumulative ranks indicating greater significance. The results are presented in Figure 3.

Fixation frequency, fixation duration (q0.75), saccade amplitude (std), and saccade velocity (median) emerge as the most influential features for predicting gender, expertise, spatial ability, and familiarity, respectively. Notably, the top five features for recognizing spatial ability are all saccade-related, underscoring the critical role of saccadic movements in spatial ability classification. For task inference, total time-related features such as FixationTestTime and

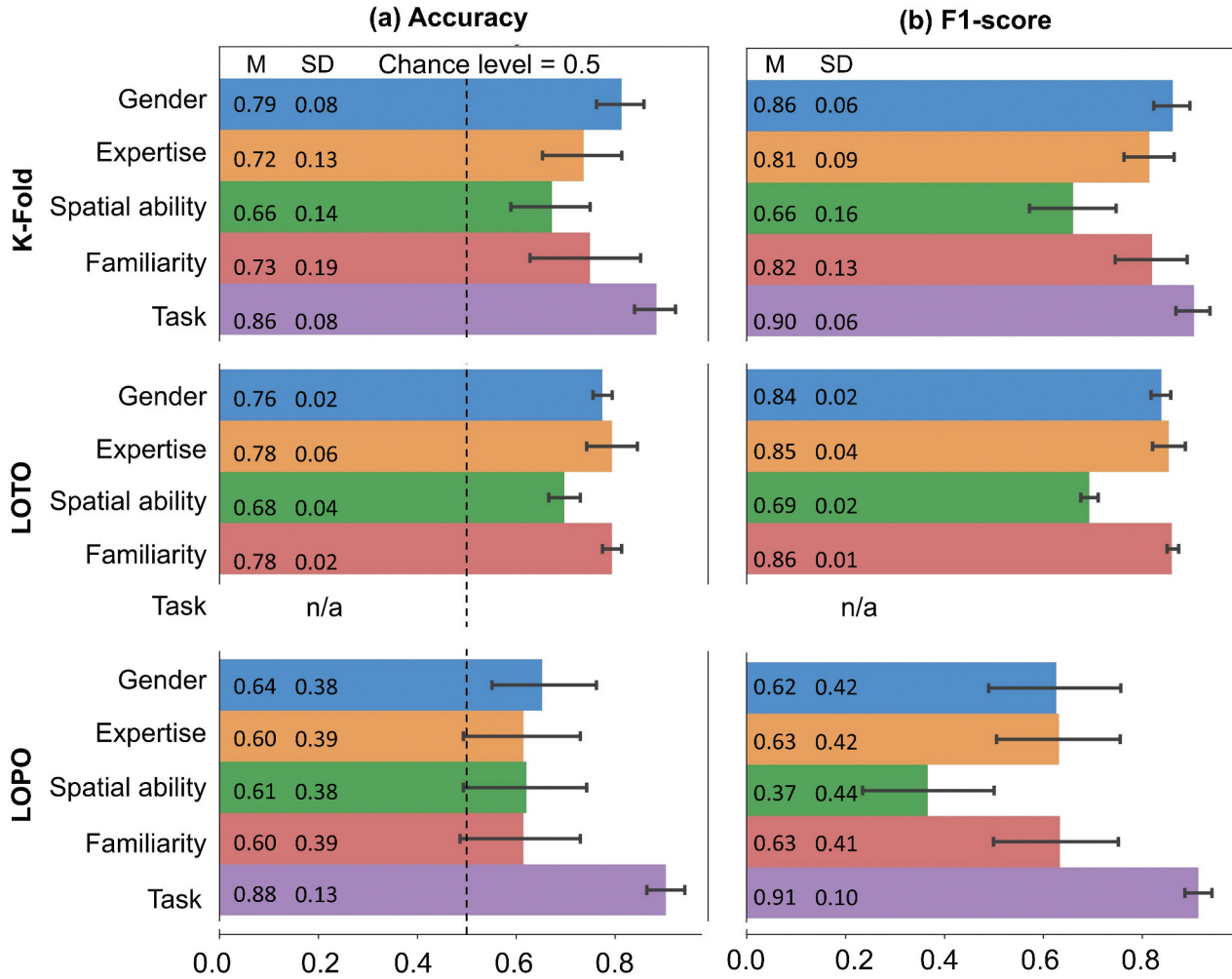


Figure 2. Accuracy and F1-score of the recognition of five user attributes across cross-validation scenarios in study 1.

SaccadeTestTime, are more influential than other features.

To further investigate how these key features differentiate between user attribute groups (e.g. males vs. females regarding fixation duration), we conducted statistical analyses to examine significant differences. Specifically, we performed: (A) normality tests and (B) tests for homogeneity of variance. Based on these results, we applied: t-tests (if data met both criteria), Welch's t-tests (if data met criterion A but not B), and Mann-Whitney U tests (if data did not meet criterion A).

Figure 4 presents the results. It is seen that significant differences in fixation-, saccade-, and/or time-related between groups were detected in each user attribute. We will elaborate on these results and discuss them in relation to the findings from Study 2 in the discussion section (Section 4).

3. Study 2: outdoor navigation in real environment

3.1. Overview

Study 2 extended user attribute recognition by incorporating a broader set of eye movement features. The data were obtained from two outdoor wayfinding experiments conducted in real-world environments by Liao et al. (2022). A total of 38 university students (17 females) from various backgrounds completed wayfinding tasks across four routes. Two of these routes are located within the campus and the other are located in residential areas. Each route is approximately 500 m. An SMI eye tracking glasses (ETG) (Apple, USA; <https://www.smivision.com>) was used to record the eye movement data (60 Hz) and forward-facing video (25 Hz). Participants were provided with a printed

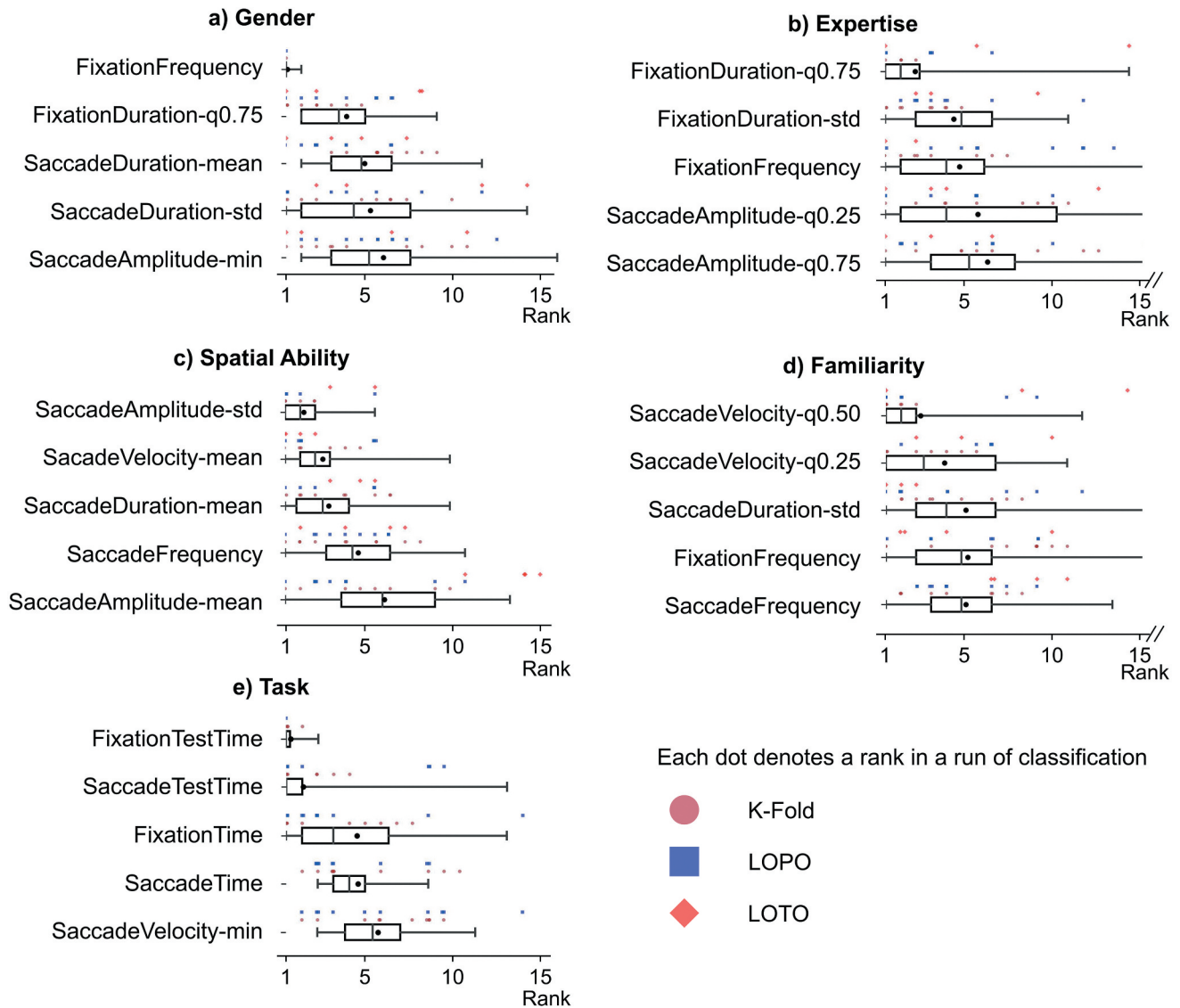


Figure 3. The top 5 features for recognition of five user attributes in Study 1.

map for reference during navigation. All participants gave written informed consent and received compensation. The experiment was reviewed and approved by the local Institutional Review Board. Further experimental details can be found in Liao et al. (2022).

We examined the same five user attributes as in Study 1.

- (1) **Gender.** The participant pool comprised 17 females and 21 males.
- (2) **Expertise.** Participants were classified into two groups based on their academic disciplines: Geo ($N = 18$) and Non-geo group ($N = 20$).
- (3) **Familiarity.** Participants rated their familiarity with each route using a 7-point scale (1 = very unfamiliar to 7 = very familiar). The mean

familiarity score across the four routes was 4.08 ($SD = 2.75$). Participants with a self-reported familiarity level < 4 were labeled “unfamiliar,” while those with familiarity > 4 were labeled “familiar.” Data points where familiarity = 4 were excluded to maintain a clear distinction. Unlike Study 1, familiarity was route-specific, meaning that each participant could have varying familiarity levels across different routes. This led to 62 trials in the familiar group and 61 trials in the unfamiliar group.

- (4) **Spatial ability.** The SBSODS was employed to measure spatial ability ($M = 44.89$, $SD = 17.57$). Participants were divided into “high” and “low” spatial ability groups based on the median score, resulting in 19 participants in each group.

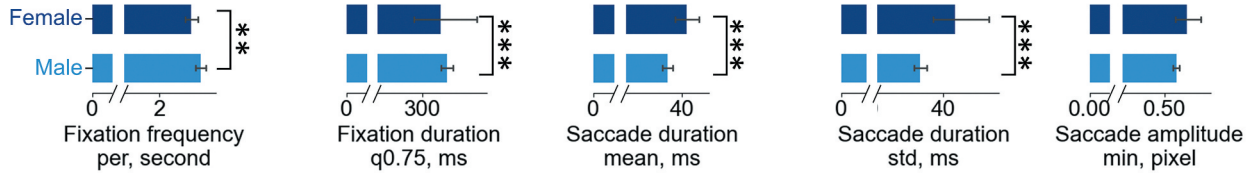
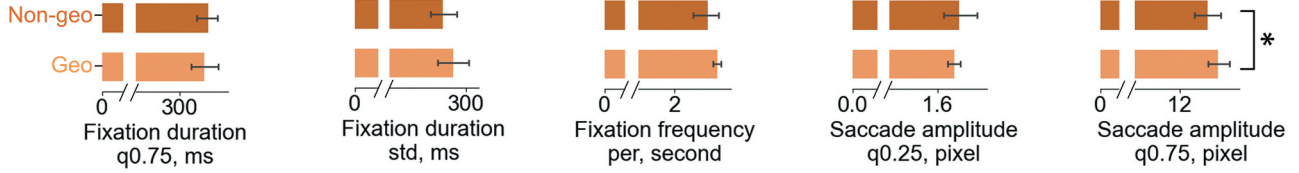
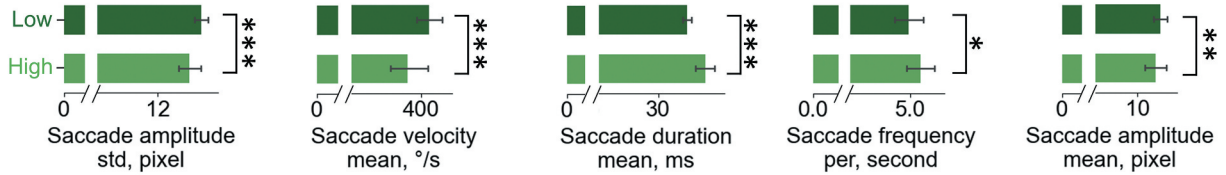
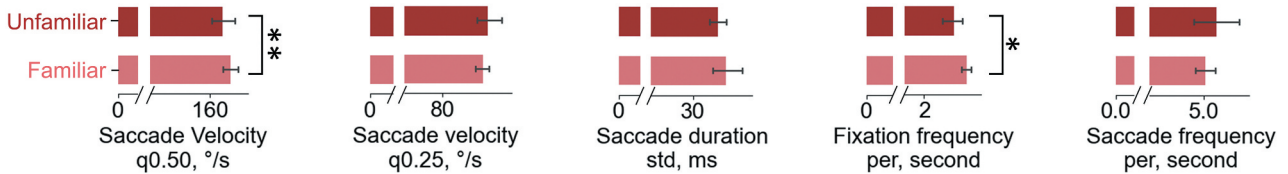
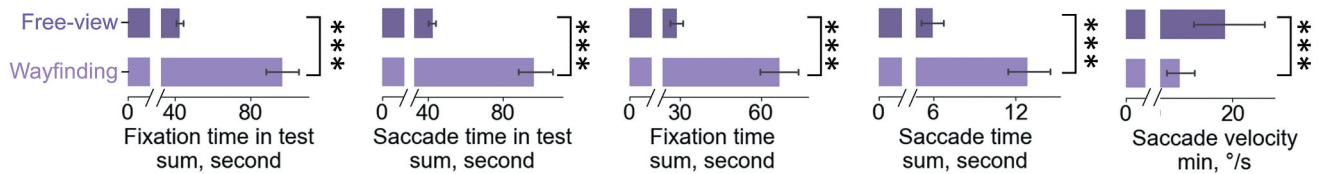
a) Gender**b) Expertise****c) Spatial ability****d) Familiarity****e) Task**

Figure 4. Statistical analysis of the top five important features in user attribute recognition in study 1. Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

- (5) **Task:** On the first two routes, participants were asked to complete five distinct tasks including determining their spatial locations and north directions (T1), finding an object in the surroundings (T2), searching for a label on the map (T3), memorizing a highlighted route on the map (T4), and wayfinding to the destination (T5). In addition, participants were required to observe the surrounding environment without specific goals for 30 seconds (free-viewing task). On the last two routes, participants performed only the wayfinding task (T5) and the free-viewing task. Unlike Study 1, task recognition in Study 2 was framed as a five-class (T1 – T5) classification problem rather than a binary classification task.

3.2. Data processing and feature extraction

The dataset comprises 146 trials, including 30.25 hours of video and 281,678 fixations. The mean duration for tasks T1 to T4 was 21.53 s (SD = 11.24 s), whereas T5 had a longer duration of 418.58 s (SD = 102.02 s), as it involved walking approximately 500 m. To maximize data utilization and examine the influence of time length on attribute recognition performance, we replicated the sliding time window method from Liao et al. (2022) to segment each trial's data. In our implementation, the window size defines the duration of each analysis segment, whereas the step size controls how far the window moves forward when generating consecutive segments.

- (1) Recognition of gender, expertise, familiarity, and spatial ability: We used only the data from T5 (four routes) and set the time window ranging from 5 s to 100 s, with 5-s increments (step = window size), resulting in 20 window sizes. Segmenting 146 trials yielded 11,996 data segments at the 5-s window and 672 segments at the 100-s window, totaling 44,719 segments.
- (2) Recognition of user tasks: We used data only from the first two routes, each containing 5 tasks, varied the time window size from 1 to 20 seconds with increasing in 1-s increments, and fixed the step of 4 seconds (step = 4 s) according to Liao, Dong, et al. (2019), resulting in 20 different window sizes. This segmentation produced 2354 data segments at a 1-s window and 2163 segments at a 20-s window, totaling 46,812 segments.

For each data segment, we extracted five types of eye movement features, resulting in a total of 581 features for 5 user attributes. Notably, the number of features in Study 2 was larger than in Study 1. These additional features were incorporated to enable the classifier to learn richer information.

3.2.1. Basic statistical features

This feature type included 8 statistical measures: mean, max, min, skewness, median, std, q0.25, and q0.75. These measures were calculated for 11 eye movement metrics based on fixation, saccade, and blink events. Specifically, fixation features were derived from fixation duration and dispersion; saccade features included duration, amplitude, velocity, latency, acceleration, acceleration/deceleration peak, and velocity peak; and blink features included duration. This resulted in $8 \times 11 = 88$ features. Additional five metrics were calculated, namely, fixation/saccade/blink frequency, scanpath length, and convex hull area of fixations, resulting in a total of 93 features in this category.

3.2.2. Pupil dilation features

In outdoor environments, pupil diameter is influenced by light intensity (Ahmad et al., 2023; Mathôt & Vilotijević, 2022). Following Mathôt and Vilotijević (2022), we used the following two baseline correction methods to minimize this influence.

- (1) **InteR task pupil dilation (IRTPD)**. For each participant and route, we calculated the mean pupil diameter of the environment free view task (PDFv) as a baseline. The IRTPD of time

t was computed by subtracting PDFv from pupil diameter (PD), as defined in Equation (3).

$$IRTPD_{p,r,t} = PD_{p,r,t} - PDFv_{p,r} \quad (3)$$

where p, r, t represents the participant, route, and time, respectively. For each data segment, we computed its 8 statistical measures of IRTPD, resulting in 8 features.

- (2) **Intra task pupil dilation (IATPD)** We used the mean pupil diameter at the beginning period of each task as a baseline. There is no consensus on the length of the baseline window, we selected 7 windows for calculation: 50 ms, 100 ms, 500 ms, 1000 ms, 2000 ms, 5000 ms and 10,000 ms. The IATPD of time t was calculated by subtracting the average pupil diameter of a given window (PDwin), as defined in Equation (4).

$$IATPD_{p,r,w,t} = PD_{p,r,w,t} - PDwin_{p,r,w} \quad (4)$$

where p, r, w, t represents the participant, route, window, and time, respectively. For each data segment and each baseline time window, we computed 8 statistical measures of IATPD, resulting in a total of $8 \times 7 = 56$ features.

3.2.3. Fixation density features

We used the Gaussian kernel density function to compute the fixation density to describe the spatial distribution of participants' visual perception (Silverman, 1986). Considering the size and information volume, a 20×20 grid is used to down-sample the screen images and stacked them into a 1×400 vector.

3.2.4. Fixation semantic features

Fixation semantic features were used to explore the uniqueness of an individual's observation of environmental objects in wayfinding. Replicating Dong et al. (2020), we used scale-invariant feature transform (SIFT) (Lowe, 2004) and the Deeplabv3+ (Chen et al., 2018) to semantically segment video frames, assigning each pixel within 20 object categories (e.g. road, person and map). The fixations were then semantically labeled by overlaying the fixations onto the segmented video frames. Finally, the fixation duration and count for each category were calculated, producing 40 features.

3.2.5. Saccade encoding features

Following (Bulling et al., 2011), we encoded saccades into character strings based on their directions (4- and 8-directions) and lengths. The strings were then divided

into substrings through a sliding window of varying length. The number of occurrences was counted as features, leading to a total of 40 features.

3.3. Classifier selection and cross-validation

Similar to Study 1, we evaluated 11 classifiers using 10-fold ($K = 10$) cross-validation (Table 2). LightGBM model achieved the highest performance across all five user attributes (all accuracies > 0.8) and was selected for subsequent analysis. For each time window and attribute, we conducted a 10-fold grid search to determine the optimal hyperparameters for model training and validation.

We employed three cross-validation methods to partition the training and testing sets: K-fold ($K = 10$), LOPO (see Section 2.5) and Leave-One-Route-Out (LORO). The purpose of LORO was to assess the model's generalizability in recognizing user attributes across new environments.

3.4. Feature selection

Given that filtering relevant features and reducing redundant information can enhance computational efficiency, learning accuracy, and model performance (Cai et al., 2018), we performed a feature selection process using cross-classifier validation. (1) *Initial Ranking*: A random forest classifier was first used to detect five user attributes and rank the importance of features for each attribute. (2) *Refining Features*: The top n important features were incrementally selected and fed into the LightGBM model for user attribute recognition.

As shown in Figure 5, both accuracy and F1-score increased rapidly when $n < 60$, but plateaued beyond $n > 100$. This trend suggests that excess features can be removed without compromising classification performance. We selected the top 100 most important features for subsequent recognition. This feature selection

strategy enhanced computational efficiency while preserving strong predictive performance.

3.5. Results

3.5.1. Performance across time windows and cross-validation scenarios

Figure 6 and Table 3 present the accuracy and F1-scores of user attribute recognition across different time windows and three cross-validation scenarios.

In the K-fold cross-validation (Figure 6a), the accuracy and F1-score remain above 0.80 for most time windows. The highest accuracy for each user attribute is as follows: gender ($ACC = 0.869$, $F1 = 0.869$ at 30 s), expertise ($ACC = 0.871$, $F1 = 0.871$ at 70 s), spatial ability ($ACC = 0.864$, $F1 = 0.863$ at 75 s), familiarity ($ACC = 0.901$, $F1 = 0.901$ at 5 s), and task ($ACC = 0.957$, $F1 = 0.957$ at 20 s). Notably, the performance for gender, expertise, spatial ability and familiarity fluctuated slightly but remained consistently above 0.80 across all time windows, indicating that classification accuracy for these attributes is relatively robust to window length.

In the LORO scenario (Figure 6b), the highest accuracies for gender, expertise, and spatial ability are 0.744 ($F1 = 0.744$ at 85 s), 0.772 ($F1 = 0.771$ at 45 s), and 0.772 ($F1 = 0.770$ at 65 s), respectively. However, familiarity recognition shows a sharp decline in performance, with accuracy ranging between 0.345 and 0.418 ($F1 = 0.459$ to 0.545), which is significantly below the chance level (0.5).

The LOPO results (Figure 6c) reveal contrasting patterns with LORO. Familiarity recognition achieves an accuracy between 0.732 and 0.821, outperforming gender (0.448 ~ 0.528), expertise (0.544 ~ 0.620), and spatial ability (0.526 ~ 0.594). Notably, the F1-scores for these attributes in LOPO remain affected by changes in time windows, suggesting sensitivity in

Table 2. The performance of 11 classifiers for recognizing five user attributes in study 2. Mean accuracy (ACC) and mean F1-score of 10-fold cross-validation were presented. The bold values represent the best.

Classifier	Gender		Expertise		Spatial Ability		Familiarity		Task	
	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
GBoost	0.777	0.777	0.804	0.803	0.797	0.796	0.844	0.843	0.868	0.865
LightGBM	0.829	0.829	0.846	0.845	0.833	0.832	0.859	0.859	0.889	0.887
MLP	0.765	0.765	0.793	0.793	0.782	0.782	0.780	0.780	0.792	0.790
AdaBoost	0.705	0.705	0.746	0.745	0.732	0.732	0.808	0.808	0.588	0.555
DecisionTree	0.660	0.661	0.685	0.685	0.679	0.679	0.718	0.718	0.793	0.794
LDA	0.624	0.636	0.630	0.630	0.643	0.643	0.636	0.636	0.741	0.740
KNN	0.766	0.766	0.802	0.801	0.769	0.768	0.744	0.744	0.723	0.714
LR	0.679	0.679	0.729	0.729	0.718	0.717	0.755	0.755	0.777	0.769
SVC	0.688	0.686	0.743	0.741	0.724	0.720	0.750	0.749	0.711	0.668
XGBoost	0.817	0.817	0.833	0.832	0.817	0.816	0.854	0.854	0.882	0.880
RF	0.785	0.785	0.808	0.808	0.795	0.793	0.829	0.827	0.868	0.863

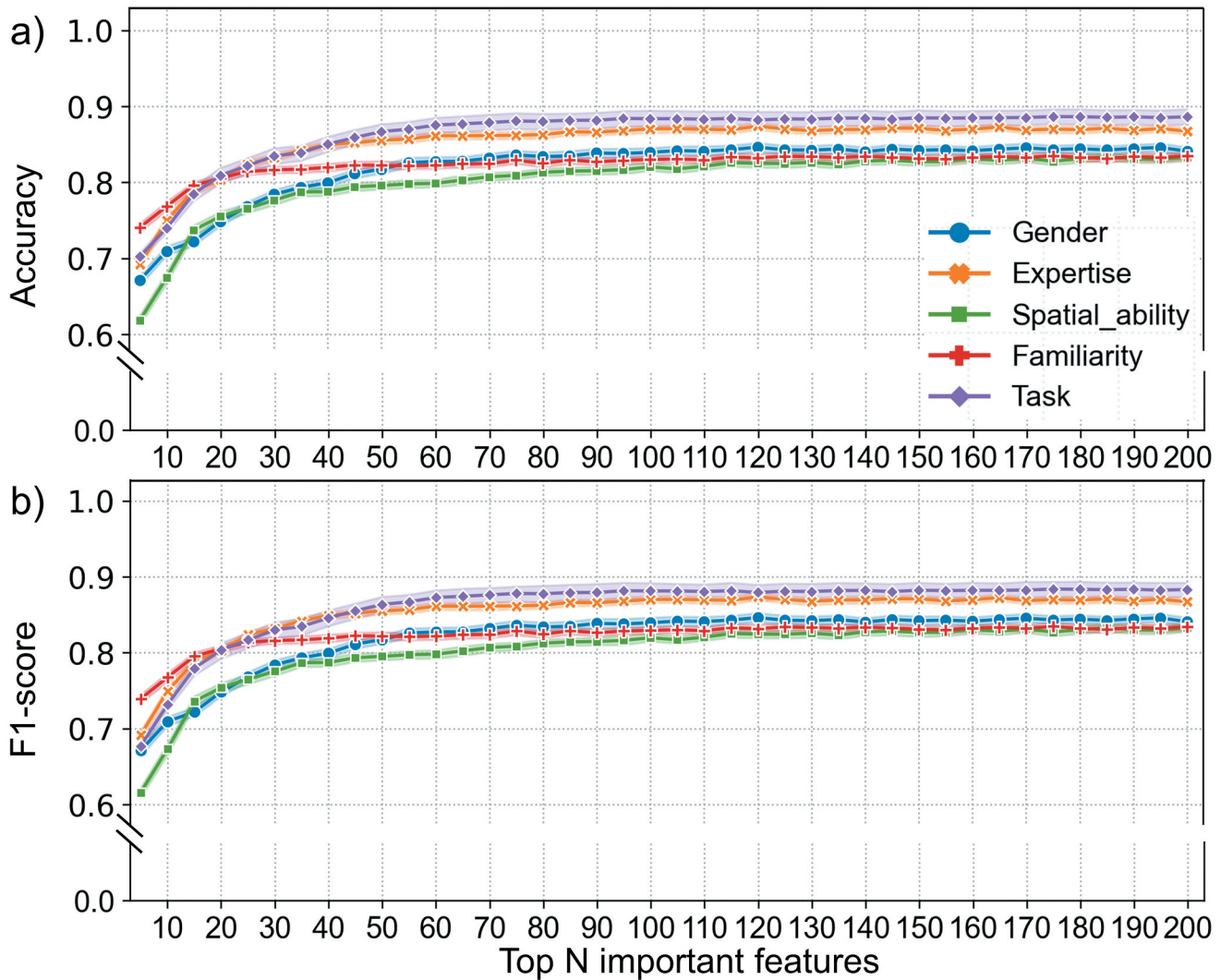


Figure 5. The performance of top n features in user attribute recognition in study 2.

the model's performance across different time windows.

Task recognition consistently outperforms other attributes across all three cross-validation methods. In K-fold, it achieves an accuracy between 0.723 and 0.957 ($F1$: 0.717 ~ 0.957), while in LORO, the accuracy ranges from 0.625 to 0.757 ($F1$: 0.625 ~ 0.738). In LOPO, task recognition attains an accuracy of 0.690 to 0.879 ($F1$: 0.684 ~ 0.874). Notably, task performance continues to improve as the time window expands, suggesting that longer observation periods contribute to more reliable task recognition.

3.5.2. Feature importance and statistical analysis

We used the comprehensive ranking method in Study 1 (Section 2.6.2) to filter the top five important features for each attribute recognition (Figure 7). Compared to Study 1, the introduction of a larger set of eye movement features resulted in notable differences in feature importance. Pupillary response features and basic statistical features,

such as blink frequency, blink duration and fixation dispersion, played an important role in recognizing gender, expertise, and spatial ability. In contrast, semantic features, such as fixation duration on maps and sidewalks, held greater importance in recognizing familiarity and task. Similar to Study 1, we conducted a statistical analysis to examine whether significant differences existed between groups for each attribute. The results are illustrated in Figure 8 and will be discussed in detail in Section 4.

4. Discussion

4.1. RQ1 (feasibility): can gaze behavior collected during pedestrian navigation in immersive VR and real-world settings reliably predict user attributes?

This study extends gaze-based user profiling beyond fixed-space controlled laboratory environments to immersive VR and RE navigation settings. Using

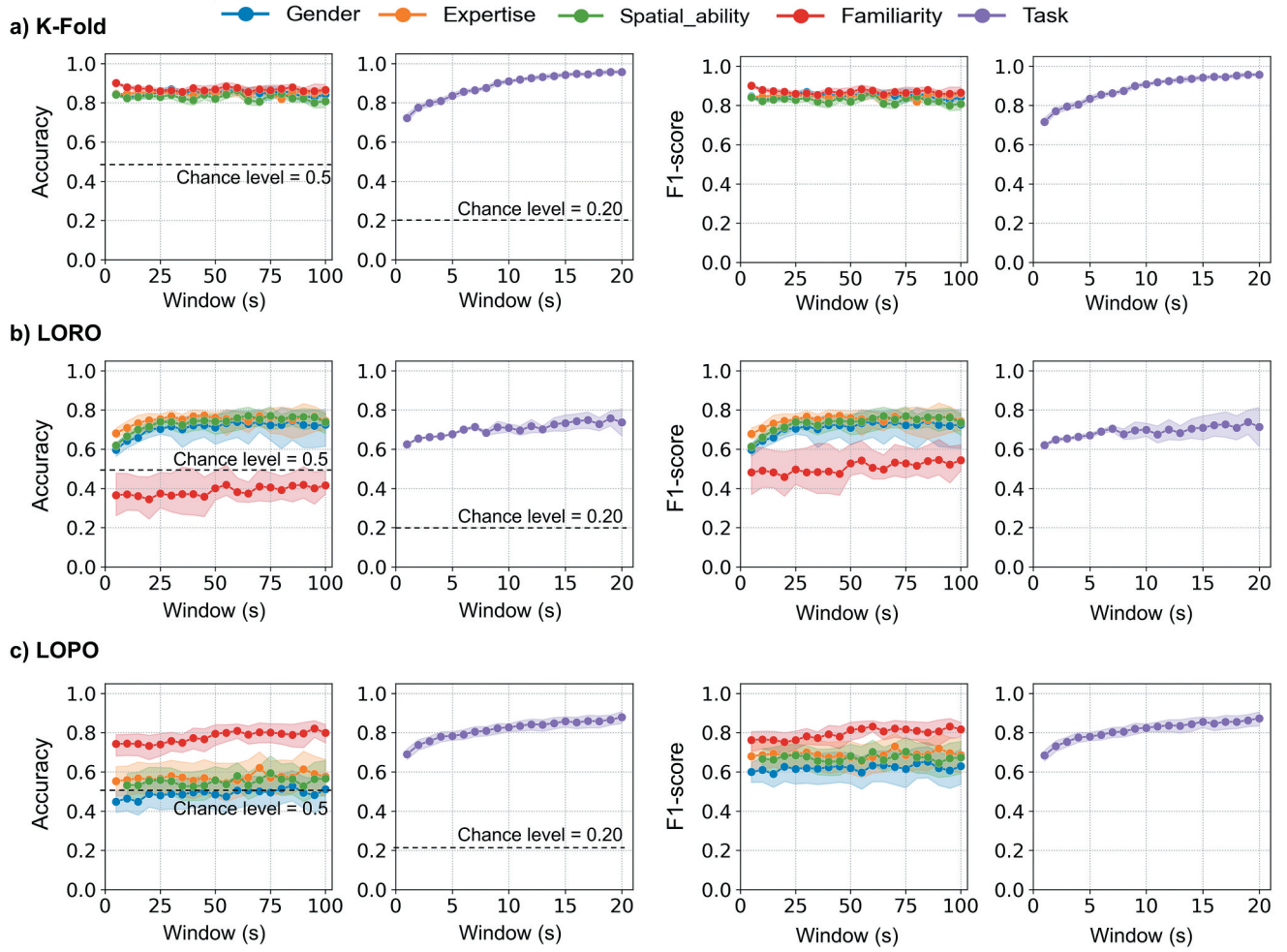


Figure 6. Accuracy and F1-score of the recognition of five user attributes across cross-validation scenarios and across time windows in study 2.

Table 3. Accuracy and F1-score in study 2.

Cross-validation	User attribute	Accuracy		F1-score	
		Range (min-max)	Mean	Range (min-max)	Mean
K-fold	Gender	0.826–0.869	0.848	0.826–0.869	0.848
	Expertise	0.820–0.871	0.851	0.820–0.871	0.850
	Spatial ability	0.801–0.864	0.828	0.800–0.863	0.827
	Familiarity	0.855–0.901	0.870	0.854–0.901	0.869
	Task	0.723–0.957	0.889	0.717–0.957	0.887
LORO	Gender	0.597–0.744	0.709	0.597–0.744	0.709
	Expertise	0.782–0.772	0.751	0.678–0.771	0.749
	Spatial ability	0.619–0.772	0.736	0.614–0.770	0.734
	Familiarity	0.345–0.418	0.385	0.459–0.545	0.507
	Task	0.625–0.757	0.705	0.625–0.738	0.691
LOPO	Gender	0.448–0.528	0.490	0.589–0.652	0.620
	Expertise	0.544–0.620	0.572	0.655–0.730	0.688
	Spatial ability	0.526–0.594	0.550	0.646–0.703	0.672
	Familiarity	0.732–0.821	0.777	0.752–0.832	0.795
	Task	0.690–0.879	0.817	0.684–0.874	0.813

K-fold validation, the VR dataset achieved peak accuracies ranging from 0.656 to 0.863 (F1: 0.659–0.903) across attributes, while the RE dataset

achieved even higher accuracies (0.864–0.957; F1: 0.863–0.957). These accuracies are comparable to or higher than existing studies despite variations in

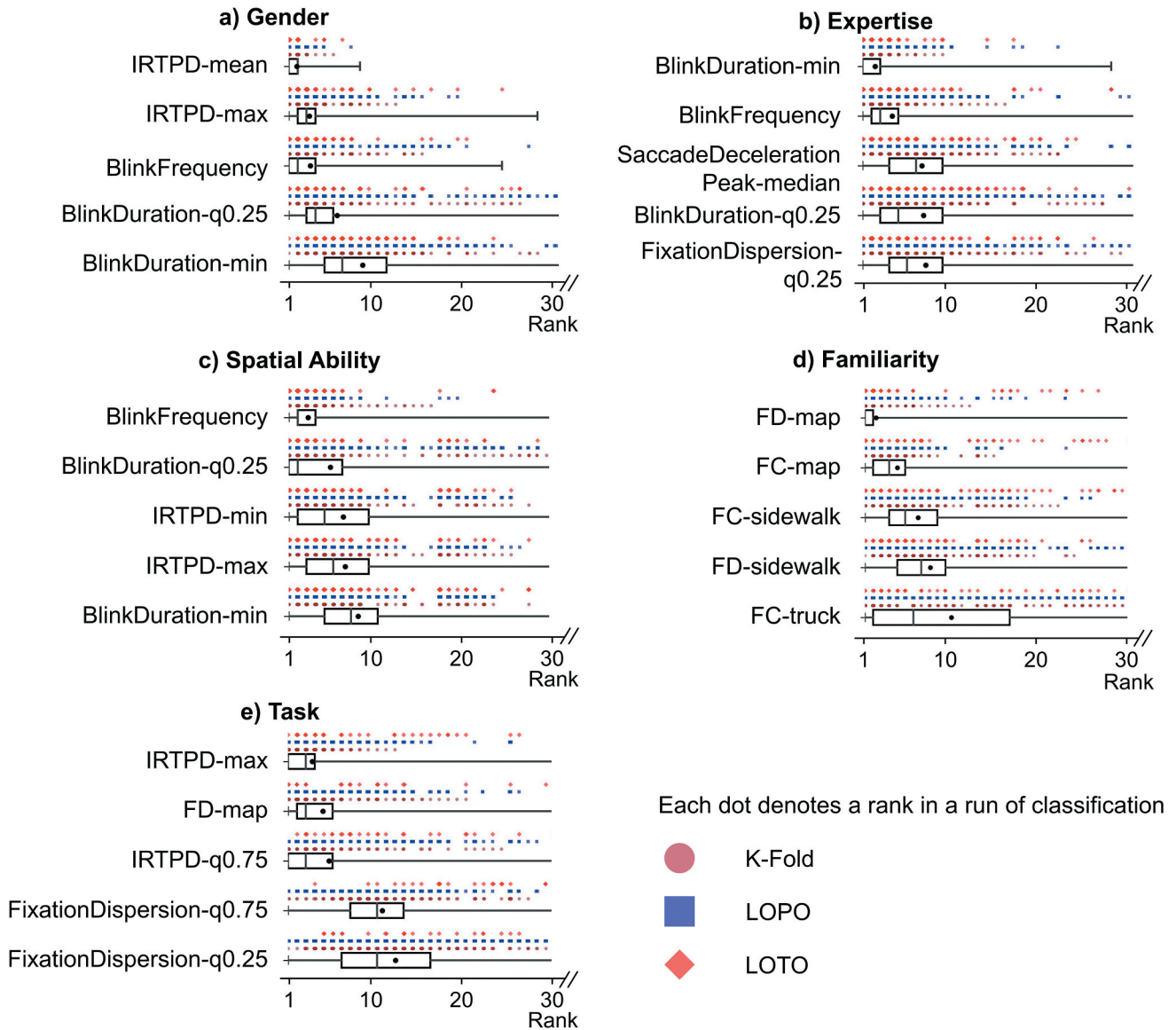


Figure 7. The top 5 features for recognition of five user attributes in Study 2.

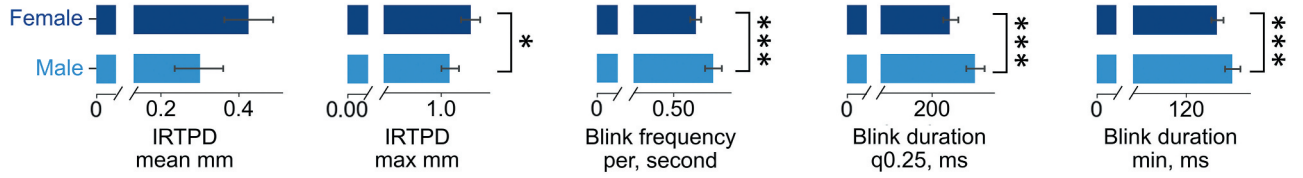
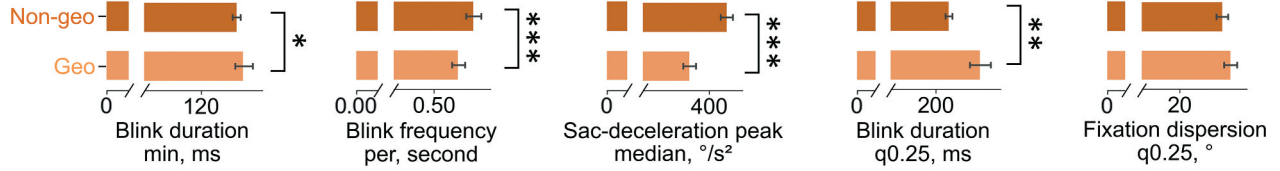
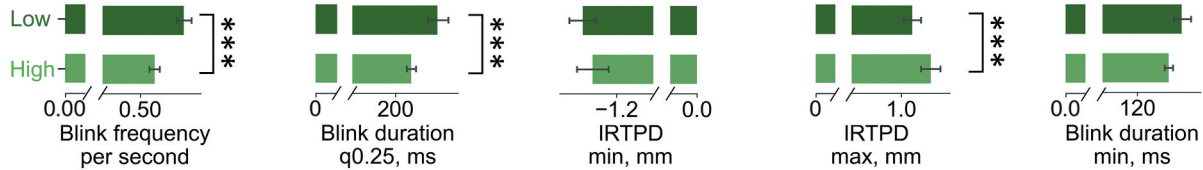
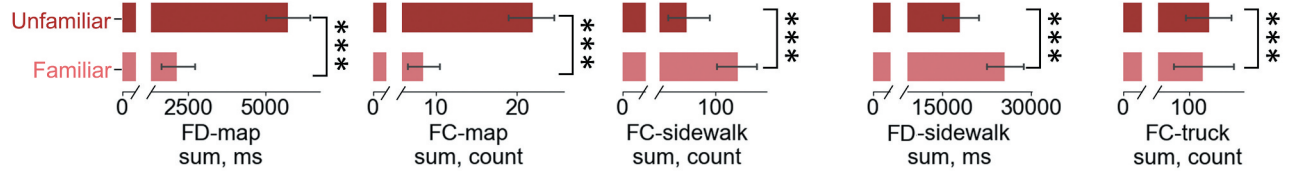
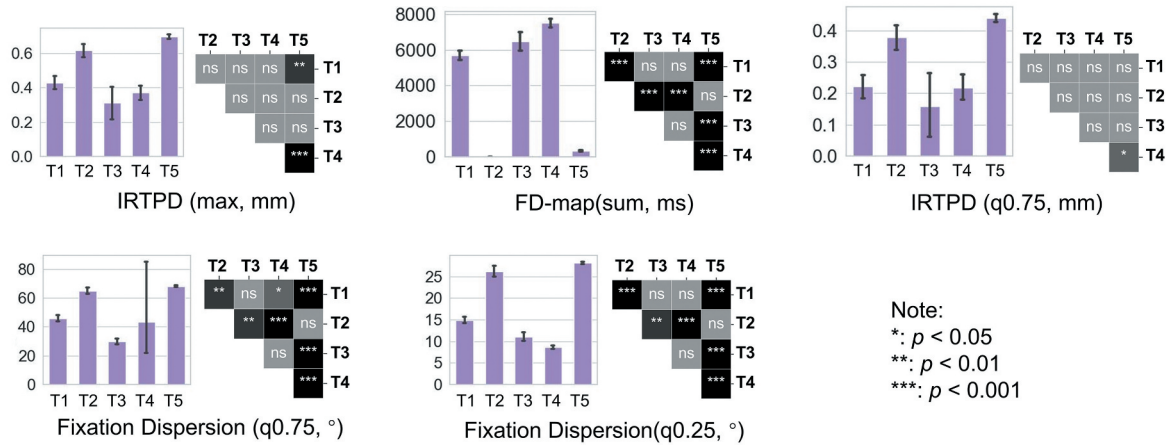
experimental environments, stimuli, and tasks (Abdi Sargezeh et al., 2019; Gokl et al., 2019; Hu et al., 2023; Kattenbeck et al., 2025; Liao, Dong, et al., 2019; Moss et al., 2012; Qin et al., 2024; Steil et al., 2019; Vallabh Varsha Haria et al., 2022). To our best knowledge, no previous studies have specifically classified users' expertise and spatial ability. The overall classification results confirm that eye movement data provide a reliable and non-intrusive basis for user profiling in both virtual and real navigation contexts.

Recognition accuracy varies across user attributes. Among the five attributes, task recognition consistently achieved the highest accuracy in both VR (Figure 2, ACC = 0.863) and RE environment (Figure 6, ACC = 0.957). This suggests that gaze patterns are strong indicators of task-related visual

and cognitive states, aligning with previous evidence (Bulling et al., 2011; Hu et al., 2023; Liao, Dong, et al., 2019; Yarus, 1967). In contrast, spatial ability and environmental familiarity were more challenging to predict, likely due to individual differences in these attributes may not produce stable gaze patterns, or the indirect manifestation of these traits in gaze metrics.

4.2. RQ2 (generalizability): how generalizable are gaze-based user recognition models across different users, tasks, and environments?

The generalizability of gaze-based user recognition models was examined using LOXO validation. While the LightGBM model achieved high accuracy under standard K-fold validation (peak accuracy:

a) Gender**b) Expertise****c) Spatial ability****d) Familiarity****e) Task**

Note:
 *: $p < 0.05$
 **: $p < 0.01$
 ***: $p < 0.001$

Figure 8. Statistical analysis of the top five important features in user attribute recognition in study 2. FD: fixation duration, FC: fixation count.

0.656–0.863 in VR and 0.864–0.957 in RE), performance decreased under LOXO scenarios, particularly when generalizing to new users (LOPO, peak accuracy: 0.600–0.881 in VR and 0.528–0.879 in RE) or unseen environments (LORO, peak accuracy: 0.418–0.772 in RE). This performance gap indicates that, although the classifier successfully learned attribute-specific gaze patterns, the model was partially dependent on the training context, limiting their transferability to unseen users or environments.

Notably, among the five user attributes, task recognition exhibited the strongest generalizability, maintaining relatively high accuracy across LOXO scenarios (peak accuracy: 0.881 in VR and 0.757–0.879 in RE). This performance is consistent with peak accuracies of 0.670 to 0.878 of previous studies using LOXO validations (Alinaghi et al., 2025; Henderson et al., 2013; Hu et al., 2023; Liao, Dong, et al., 2019; Steil et al., 2019). These results indicate that task-related gaze patterns exhibit higher

consistency across users and environments compared with other user attributes. In contrast, recognition performance for gender, expertise, spatial ability, and familiarity dropped more substantially in cross-user and cross-environment tests. This shows that these attributes are associated with more individualized or context-specific gaze behavior, which will be elaborated in Section 4.3.

4.3. RQ3 (interpretability): which eye movement features contribute most to user attribute recognition, and how do they explain cognitive differences among users?

To improve the interpretability of the classification model, we analyzed the contribution of individual eye movement features and examined how these features relate to cognitive and behavioral differences among users. The results revealed that no single feature type dominated across all attributes; rather, the most informative features varied depending on the attribute and the validation method. Generally, basic statistical features (e.g. fixation frequency, saccade duration, and velocity) and pupillary response features emerged as reliable predictors, while fixation semantic features (e.g. fixation duration on map and sidewalk) ranked highly in feature importance when predicting familiarity and task, indicating their significance in the RE environment.

4.3.1. Gender

For gender recognition, the feature importance analysis revealed that fixation-, saccade-based, and pupil dilation features played the most discriminative roles in the classifier. Specifically, shorter saccade durations (mean and std, $p < 0.001$), higher fixation frequency ($p < 0.01$; Figure 4a), and smaller IRTPD (max, $p < 0.05$; Figure 8a) were key predictors distinguishing male from female participants. These patterns align with prior findings linking increased fixation frequency and reduced saccade duration to enhanced visual processing and search efficiency (Liao, Wang, et al., 2019; Ooms et al., 2012), indicating that males processed and analyzed visual cues more rapidly than females. The smaller IRTPD suggests lower cognitive load for males (Liao et al., 2025; Mathôt et al., 2018).

From an interpretability perspective, these results show that gender classification in gaze-based profiling is driven by combinations of attention allocation and cognitive effort. Moreover, our findings are consistent with Zhou et al. (2020), who found that males showed higher fixation and saccade frequencies during wayfinding in subway stations. Their study further proposed that males tend to adopt a more exploratory or

adventurous navigation strategy, which corresponds with our observation of higher fixation frequency and lower IRTPD among male participants.

4.3.2. Expertise

For expertise recognition, the most informative features were saccade amplitude and blink-related metrics (Figures 4b and 8b). Geography participants showed significantly greater visual search extent (saccade amplitude $q_{0.75}$, $p < 0.01$) and increased fixation dispersion ($q_{0.25}$, $p > 0.05$), suggesting more extensive visual exploration to acquire global spatial information. The classifier identified shorter blink duration ($p < 0.05$) and higher blink frequency ($p < 0.001$) as features distinguishing Non-geo participants. These features suggest that non-geography participants had more frequent attentional interruptions and higher cognitive load (Biondi et al., 2023; DiStasi et al., 2011).

Previous studies indicated significant differences between experts and non-experts with experts exhibiting higher fixation frequency and shorter fixation duration in map reading tasks (Dong et al., 2018; Keskin et al., 2020; Ooms et al., 2012). However, these findings have been partially confirmed in the VR study, as differences in fixation duration ($q_{0.75}$, $p > 0.05$) and frequency ($p > 0.05$) between the groups were not significant (Figure 4b).

4.3.3. Spatial ability

Statistical results indicate that the classifier distinguished spatial ability primarily through saccade dynamics and pupillary response indicators. In the VR study (Figure 4c), participants with higher spatial ability exhibited greater saccade frequency ($p < 0.05$) and longer saccade durations (mean, $p < 0.001$) than those with lower spatial ability, yet showed shorter saccade amplitudes (mean and SD, $p < 0.01$) and lower saccade velocity ($p < 0.001$). These patterns suggest that high-ability participants engaged in more frequent, prolonged saccades covering shorter distances, potentially reflecting a goal-directed strategy that emphasizes incremental, precise visual scanning to build detailed spatial representations. However, this interpretation remains speculative, as no supporting evidence was found in fixation-based features.

In the RE study (Figure 8c), individuals with low spatial ability exhibited significantly higher blink frequency ($p < 0.001$) and longer blink duration ($q_{0.25}$, $p < 0.001$), indicating increased cognitive load (Biondi et al., 2023; DiStasi et al., 2011). However, individuals with high spatial ability exhibited larger IRTPD (max, $p < 0.001$), indicating greater pupil dilation relative to baseline. This is unexpected, as increased IRTPD typically associated with higher cognitive load (Liao et al., 2025; Mathôt et al., 2018). We interpret this as evidence

of greater task engagement or attentional allocation. Rather than indicating mental strain, the increased IRTPD may reflect active cognitive effort in processing spatial information.

4.3.4. Environment familiarity

In VR, participants showed significantly higher saccade velocity (median, $p < 0.01$) and increased fixation frequency ($p < 0.05$) in familiar environments (Figure 4d), likely reflecting proactive gaze shifts guided by prior spatial knowledge. Familiar participants could anticipate key scene elements, enabling faster and more efficient visual processing. This interpretation is supported by findings in the RE setting: as shown in Figure 8d), familiar participants checked the map less frequently (FC-map, $p < 0.001$) and for shorter durations (FD-map, $p < 0.001$) than unfamiliar participants, having already formed a mental representation of the space. Consequently, their gaze was more often directed toward the sidewalk (FC-sidewalk sum and FD-sidewalk sum, $ps < 0.001$), suggesting an attentional shift toward walking safety and obstacle avoidance rather than reorientation.

4.3.5. User task

The task recognition accuracy in RE is significantly higher than the 0.67 accuracy reported by Liao, Dong, et al. (2019). The improvement can be attributed to fixation semantic features, such as fixation duration on the map (FD-map sum). Tasks involving spatial interpretation (e.g. T1, T3, and T4) showed significantly longer map fixations ($ps < 0.001$). These features combine gaze data with spatial layout information, highlighting the value of context-aware gaze metrics in task recognition.

Additional statistical features, such as fixation dispersion (min and q0.25), further contributed to task differentiation. For example, T2 (finding environment target) and T5 (wayfinding) required longer saccade durations (q0.50 and q0.75, $ps < 0.001$) and greater fixation dispersion (min and q0.25, $p < 0.001$) than other tasks. Moreover, inter-recorded task pupil diameter (IRTPD) was significantly higher during T5 than during T1 and T4, suggesting elevated cognitive load during wayfinding.

The robust performance of task recognition across all LOXO validations is likely driven intrinsic cognitive and perceptual distinctions between tasks. Task-related gaze patterns appear largely independent of gender, expertise, spatial ability, and familiarity. Because the model was trained on samples spanning all task types, the substantial behavioral variability induced by task demands may attenuate the discriminative signals associated with other attributes. Future work employing task-controlled experimental designs or developing

task-specific attribute recognition models may help disentangle these overlapping influences.

In addition, it is worth noting that task recognition accuracy consistently increased with time window size (Figure 6). However, as the dataset limited time windows to 20 seconds, it remains unclear whether accuracy would continue to rise or plateau, which is observed with other attributes beyond 25 seconds.

4.3.6. Summary of feature interpretability across user attributes

Across all attributes, several consistent features emerged. Basic statistical features such as fixation frequency, fixation duration, saccade duration, saccade amplitude, and blink duration, proved to be broadly informative and frequently ranked among the top predictors. Pupillary response measures such as IRTPD also contributed to multiple attributes (i.e. gender, spatial ability, and task), reflecting their sensitivity to cognitive load. These findings suggest that basic eye movement features and pupil dilation-based indicators capture general aspects of users' visual search and processing, forming a shared foundation for attribute recognition. In contrast, fixation density features and saccade encoding features indicated limited relevance for distinguishing user attributes.

Other features showed attribute-specific importance. For instance, gaze semantic features were informative primarily when the attribute was tied to environment-dependent visual behavior (e.g. user task). Map-related fixations were particularly important for familiarity recognition, whereas they played minimal roles in gender or expertise prediction. These results highlight that effective gaze-based profiling relies on combining general eye movement indicators with context-sensitive semantic features to capture both shared and attribute-specific cognition patterns.

4.4. Limitations

Three additional issues should be noted. First, the generalizability of the models in this study is reflected in their environment-, individual-, and task-independence within a single experimental condition. We did not analyze the generalizability of models across different experimental conditions (i.e. whether models trained in VR can be applied to real-world data, and vice versa). This was mainly due to significant differences in experimental design and technical issues in data collection in the VR study. Future studies should address this limitation by improving experimental design and retraining models on common feature sets to test cross-context (VR versus RE) transferability.

Second, while both the VR and RE studies shared a core set of metrics (basic statistical features), the RE study also included fixation semantic features that captured contextual aspects of navigation behavior. These differences can partly explain the variation in feature importance between the two studies. Rather than indicating inconsistency, this divergence highlights their complementarity: core gaze dynamics remain robust indicators across contexts, while context-aware features enhance interpretability and classification performance in real-world scenarios.

Finally, multi-class prediction was used only for task recognition in the real-world environment and other attributes were binarized, as the relatively limited sample size in each experimental condition constrained the feasibility of multi-level classification. While binary classification enhances robustness and is a common practice (refer to Table A1), it also simplifies group distinctions. For example, environmental familiarity was measured on a 7-likert scale that cannot be accurately reduced to “familiar” versus “unfamiliar.” Employing multi-level classification for certain user attributes may yield finer distinctions, offering a valuable direction for future research.

5. Conclusions and future work

This study investigated the feasibility, generalizability, and interpretability of gaze-based user profiling during pedestrian navigation in both immersive virtual reality and real-world environments. Using a LightGBM classifier, five common user attributes were recognized based on diverse eye movement features.

- (1) Regarding feasibility, the results demonstrated that user attributes can be reliably predicted from gaze behavior using K-fold cross-validation. Task recognition consistently achieved the highest accuracy (up to 0.941), while other attributes showed strong to moderate performance in this validation scenario. These findings confirm that gaze features carry meaningful behavioral and cognitive signatures beyond controlled laboratory settings.
- (2) We used leave-one-X-out validations to assess model generalizability. The results indicate that task recognition is more generalizable than other user attributes. While K-fold cross-validation yielded high accuracy, recognition performance declined notably when generalizing to new users or unseen environments, particularly for familiarity recognition across environments.
- (3) Regarding interpretability, feature importance and statistical analyses revealed that basic statistical features (e.g. fixation frequency, saccade

duration), pupillary responses, and fixation semantic features (e.g. fixation duration on map or sidewalk) contributed most to classification. These findings provide interpretable links between gaze behavior and inferred attributes, representing an important step toward explainable and personalized navigation aids.

Overall, the results suggest that while gaze-based profiling holds promise, further improvements in model robustness and generalizability are necessary. Future research should focus on enhancing cross-context robustness, incorporating lightweight multimodal behavioral metrics (e.g. head movements), and extending experiments to larger populations and diverse navigation contexts. While physiological signals such as EEG may deepen theoretical insights, their use is likely limited to controlled environments rather than real-world deployment. Future work should therefore carefully weigh the trade-off between richer data and practical applicability.

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Data availability statement

The data and code that support the findings of this study are available at figshare.com: <https://figshare.com/s/5b06a12beae19a479aab>.

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Appendix A

Table A1. Selected studies of user attribute recognition using eye movement data.

Reference	N (#female)	Classifier	Performance (optimal)	Number of classes	Cross-validation method	Experimental environment	Experimental task
Gender							
Haria et al. (2022)	370(185)	LR RF	Accuracy: LR 65.2%, RF 64%	2	50 runs	Lab	Looking at the images of a single actor gazing toward the subjects
Abdi Sargezeh et al. (2019)	45(20)	SVM	Accuracy: 70%	2	5-fold	Lab	Watching the indoor images which contained a lot of objects
Al Zaidawi et al. (2020)	38(19)	LR	Accuracy: Non-dyslexic participants: 63.8% Dyslexic participants: 60.7%	2	1000 runs Training set: testing set = 4:1	Lab	Watching a text consisting of 10 sentences with the recording of 20s
Moss et al. (2012)	52(26)	LR	Accuracy: 64% using romantic and active images	2	10-fold	Lab	Watching and evaluating the images which selected by a particular genre of films
Steil et al. (2019)	20(10)	SVM	Accuracy: 70% with no-difference	2	Leave-one-participant-out	Lab	Reading the documents that contain images and text of varying proportions
Gao et al. (2023)	280(140)	LightGBM	Accuracy: 70.8% for best parameters	2	Nested cross-validation	Immersive VR	Conducting a virtual lesson about basic CT principles
Galdi et al. (2016)	112(39)	AdaBoost SVM	Accuracy: AdaBoost 50.85%, SVM 55.56%	2	k-fold	Lab	Looking at a sequence of 16 human face photographs
Gil-López et al. (2023)	57(27)	SVM	Accuracy: 77% for combined tasks	2	5-fold	Immersive VR	Purchasing in a immersive store
Cognitive State							
Eivazi and Bednarik (2010)	12(3)	SVM	Accuracy: 75.84%	5	3-fold	Lab	Solving an 8- tiles puzzle game which had to think aloud
Celiktutan and Demiris (2019)	31(10)	SVM	Accuracy: SVM 59.1%, GP 57.2%	2	Leave-one-participant-out	Lab	Playing trivia games with questions of different difficulty levels
Orlosky et al. (2019)	16	SVM	Accuracy: 62.8%	3		Immersive VR	Determining if the participants can understand the meaning of the words presented
Barral et al. (2020)	56(32)	XGBoost	Accuracy: 72% for predicting VerWM	2	10-fold	Lab	Reading 15 MSNVs and answering 3 comprehension questions after each trail
Environment Familiarity							
Liao et al. (2022)	38(19)	RF	Accuracy: 81%	2	10-fold Leave-one-task-out	Real-world	Completing 5 tasks each route, including self-localization, object searching, route memorization et al.
Kattenbeck et al. (2025)	52(25)	XGBoost	Accuracy: 67.6% for eye tracking data 89.9% for IMU data	2	80% for train Leave-5-out-for-testing	Real-world	Navigating familiar and unfamiliar routes
Task							
Qin et al. (2024)	50	LightGBM	Accuracy: ET: 85.9% ET+EEG: 88.0%	4	5-fold	Lab	Performing 4 daily map use tasks including global search, distance comparison, route following and route planning
Li et al. (2020)	88(44)	SVM	Accuracy: 72.87% for the questions which have not seen by the classifier	2 and 3	10-fold	Lab	Completing 2 types of questions about spatial visualization
Liao, Dong, et al. (2019)	44(20)	RF	Accuracy: 67% for combined features in 17s	5	Leave-one-participant-out	Real-world	Finishing the tasks about navigation
Henderson et al. (2013)	12	NB	Accuracy: 68% when training and testing use data at different times	4	Leave-one-out	Lab	Completing 4 tasks: memory task, reading texts, searching letters and pseudo-reading

(Continued)

Table A1. (Continued).

Reference	N (#female)	Classifier	Performance (optimal)	Number of classes	Cross-validation method	Experimental environment	Experimental task
Boisvert and Bruce (2016)	19	RF	Accuracy: 69.6%	4	hold-out for 50%	Lab	Each participant performed 1 of 3 viewing tasks or an explicit judgment task
Alinaghi et al. (2025)	56(22)	XGBoost	Accuracy: 87.8%	2 3	Leave-k-Group- Out	Real-world	Categorizing the three navigating steps
Kiefer et al. (2013)	17(10)	SVM	Accuracy: 77.7%	6	10-fold	Lab	Performing 6 map reading tasks: free viewing, global search, route planning, focused search, line following and polygon comparison

Notes: LR = logistic regression; RF = random forest; SVM = support vector machine; GP = gaussian processes; NB = Naive Bayes; LightGBM: light gradient-boosting machine.